AGNI COLLEGE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE MINI PROJECT REPORT

SMART AGRI-CROP RECOMMENDATION SYSTEM

A Data-Driven Decision Support System Prototype for Precision Agriculture
Evaluating 7 Soil & Climatic Parameters Across 18 Crop Profiles

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ABSTRACT

Modern agriculture faces critical challenges due to traditional crop selection practices, which rely heavily on historical intuition and regional habits rather than empirical soil chemical testing and real-time microclimate observations. Improper crop selection leads to land nutrient depletion, unoptimized fertilizer consumption, lowered harvest yields, and severe financial risk under fluctuating weather conditions.

This mini-project report presents the design and prototype implementation of the Smart Agri-Crop Recommendation System, an interactive web-based decision support platform built using React 19, TypeScript, Vite, Tailwind CSS, and Lucide React. The platform evaluates seven core agricultural input parameters: Nitrogen (N), Phosphorus (P), Potassium (K), Soil pH, Temperature, Humidity, and Rainfall.

The prototype incorporates a client-side agronomic distance scoring engine (`calculateCropRecommendations` in `src/utils/recommendationEngine.ts`) that processes feature vectors against a dataset of 18 crop varieties. It calculates continuous weighted suitability percentages and ranks the primary recommended crop alongside the Top 3 alternative crop varieties. Additionally, the project establishes a modular architecture for planned future integration with a server-side

Python Flask REST API (`POST /api/predict` in `backend/app.py`) for machine learning models (`model.pkl`).

This report documents the problem statement, objectives, system architecture, component design, prototype user interface, algorithm workflow, and prototype testing observations, serving as a comprehensive, faculty-viva verified submission.

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CHAPTER 1: INTRODUCTION

1.1 Background of Agriculture and Technology Integration

Agriculture remains the primary foundation of food security, rural employment, and economic stability in India. However, traditional agricultural methodologies rely heavily on subjective decision-making, historical regional practices, and unverified trial-and-error techniques. Farmers often cultivate specific crops based on previous season outcomes or neighbor recommendations without evaluating current soil chemistry or seasonal microclimate predictions.

With the advent of computer science, modern web technologies, and computational data processing, precision agriculture has emerged as a crucial domain. Precision agriculture leverages soil chemistry measurements, microclimate statistics, and automated decision algorithms to optimize crop yield, lower resource wastage, and protect long-term soil health.

1.2 Importance of Soil and Environmental Vector Analytics

Crop growth depends on complex biochemical and physical interactions between soil nutrients and environmental conditions. Nitrogen (N), Phosphorus (P), and Potassium (K) serve as primary macronutrients governing foliage synthesis, root expansion, and plant immunity. Soil pH dictates nutrient bioavailability to root systems, while climatic variables—temperature, humidity, and rainfall—regulate photosynthetic respiration and moisture uptake.

When any single parameter deviates significantly from optimal crop growth thresholds, plant development suffers, resulting in stunted growth, pest susceptibility, or complete yield failure. Analyzing these seven parameters simultaneously as a multi-dimensional feature vector enables accurate compatibility evaluation.

1.3 Project Scope and System Overview

The Smart Agri-Crop Recommendation System mini-project aims to bridge the gap between complex agricultural science and modern web interfaces. The project delivers a functional, responsive web application prototype built using React 19, TypeScript, Vite, Tailwind CSS, and Lucide React.

The current prototype features an intuitive frontend user interface where farmers can input numerical soil and environmental metrics or select quick test presets (e.g., Clay/Rice, Black Soil/Cotton). A deterministic client-side recommendation engine (`calculateCropRecommendations`) evaluates inputs against 18 crop varieties in `cropDataset`, providing instant suitability scores, parameter breakdown tables, and alternative crop choices. Additionally, the system defines a modular application architecture connecting to a Python Flask REST backend (`POST /api/predict` in `backend/app.py`) for planned future machine learning inference.

CHAPTER 2: PROBLEM IDENTIFICATION AND OBJECTIVES

2.1 Problem Statement

Traditional crop selection methods suffer from critical structural deficiencies that directly impact agricultural productivity and soil sustainability:

- **Guesswork and Legacy Practices:** Farmers select crops based on historical memory or regional habit without conducting empirical soil nutrient or climatic suitability assessments.
- **Soil Degradation and Over-Fertilization:** Excessive or improper fertilizer application disrupts ground pH balance, causing land degradation and waste of capital.
- **Climate Change Vulnerability:** Unpredictable thermal changes and erratic rainfall lead to crop mortality when crops mismatched to microclimates are planted.
- **Complex Soil Reports:** While soil testing laboratories provide numerical laboratory reports (e.g., Nitrogen = 90 mg/kg), farmers lack accessible tools to map raw data into actionable crop choices.

2.2 Project Objectives

To address the identified problems, the Smart Agri-Crop Recommendation System sets forth the following primary objectives:

- **1. Feature Vector Ingestion:** Design a clean web form capable of collecting seven agricultural features (Nitrogen, Phosphorus, Potassium, Soil pH, Temperature, Humidity, and Rainfall) with active range validation.
- **2. Scientific Compatibility Scoring:** Implement a multi-attribute distance engine that evaluates soil chemistry and climate metrics against 18 crop dataset profiles.
- **3. Interactive Web Prototype:** Construct a responsive web user interface (React 19 + TypeScript + Vite) offering test presets, interactive input sliders, parameter status badges, and top 3 crop rankings.
- **4. Modular Backend Architecture:** Establish a server-side architecture (Python Flask REST API at POST /api/predict) capable of serving trained machine learning models (model.pkl) for proposed server-side predictions.

CHAPTER 3: EXISTING SYSTEM AND PROPOSED SYSTEM

3.1 Existing System Overview

The existing agricultural advisory ecosystem in many rural areas relies on manual consultations, static printed literature, or single-attribute lookup tables. Soil testing centers issue physical paper reports containing numerical concentrations of soil nutrients. Farmers must interpret these numbers independently or seek verbal guidance from local extension officers.

Disadvantages of Existing Systems:

- **Lack of Automation:** Subjective advice based on personal opinion rather than quantitative mathematical scoring.
- **Single-Parameter Scope:** Focuses only on soil nutrients while ignoring temperature, humidity, and seasonal rainfall.
- **No Interactive Real-Time Feedback:** Static paper reports cannot simulate 'what-if' weather scenarios or adjust recommendations dynamically.
- **No Alternative Ranking:** Provides a single crop suggestion without secondary options if market or climatic conditions change.

3.2 Proposed System Overview

The proposed Smart Agri-Crop Recommendation System introduces a data-driven digital platform. It combines soil chemical parameters and climatic metrics into a

unified feature vector, evaluating multi-dimensional compatibility instantly through an interactive web prototype.

3.3 Comparative Matrix

Comparison Dimension	Existing System / Practice	Proposed Smart Agri System
Decision Basis	Manual intuition & regional habit	Quantitative empirical feature analytics
Feature Vector	Single attribute (soil only)	7-Vector multi-attribute fusion (N-P-K-pH-Temp-Hum-Rain)
User Interface	Static paper reports	Interactive React web prototype dashboard
Output Depth	Single binary advice	Primary recommended crop + Compatibility % + Top 3 alternatives
System Architecture	Offline manual consults	Decoupled React Frontend + Python Flask REST API

Table 3.1: Comparative Matrix between Existing System and Proposed System

CHAPTER 4: SYSTEM REQUIREMENTS AND TECHNOLOGIES USED

4.1 Hardware and Software Requirements

Category	Requirement Component	Specification
Hardware (Development)	Processor & RAM	Intel Core i5 / AMD Ryzen 5, 8 GB RAM
Hardware (Client)	Display & Input	1024x768 Resolution, Keyboard/Mouse/Touch screen
Operating System	Environment OS	Windows 10 / 11, Linux, or macOS
Frontend Runtime	Web Browser & Node.js	Modern Chrome/Edge/Firefox, Node.js v18+
Backend Environment	Python Environment	Python 3.10+ (Flask, Scikit-learn, Pandas, NumPy)

4.2 Technologies Used

- **React 19:** React 19.2.8 is used as the core frontend UI framework, providing declarative component rendering.
- **TypeScript:** TypeScript is used across the codebase to ensure type safety for input vectors, crops, and result types.
- **Vite:** Vite 8.3.0 is configured as the dev server and module bundler with fast Hot Module Replacement.
- **Tailwind CSS:** Tailwind CSS 4.3.3 (`@tailwindcss/vite`) is imported in `index.css` for visual styling.
- **Lucide React:** Lucide React 1.49.0 provides SVG agricultural and UI icons.
- **Framer Motion:** Framer Motion 13.5.0 drives smooth tab transitions and result card animations.
- **Python & Flask Backend:** Python 3 script (`backend/app.py`) uses Flask and Flask-CORS to define prototype API route `POST /api/predict`.
- **Scikit-Learn, Pandas, NumPy:** Scikit-Learn, Pandas, and NumPy are specified in `backend/requirements.txt` for planned server-side model loading.
- **VS Code, Git & GitHub:** VS Code, Git, and GitHub were utilized for version control and development.

CHAPTER 5: METHODOLOGY AND INPUT PARAMETERS

5.1 7 Core Input Parameters

The system processes a 7-element feature vector representing key soil nutrients and climatic metrics:

Parameter	Unit	Category	Agronomic Role & Growth Impact
Nitrogen (N)	mg/kg	Soil Nutrient	Drives foliage growth, leaf development, and chlorophyll synthesis.
Phosphorus (P)	mg/kg	Soil Nutrient	Stimulates root expansion, early seed development, and flowering.
Potassium (K)	mg/kg	Soil Nutrient	Improves plant disease immunity,

			stomatal regulation, and drought resilience.
Soil pH	pH (0-14)	Soil Chemical	Measures acidity/alkalinity, governing root nutrient bioavailability.
Temperature	°C	Microclimate	Thermal growth window regulator governing cellular respiration rate.
Humidity	Relative %	Microclimate	Atmospheric moisture level influencing leaf transpiration balance.
Rainfall	mm / season	Hydrological	Precipitation volume satisfying seasonal crop water requirement.

5.2 Verified Suitability Scoring Formula (Exact Implementation)

In the frontend engine prototype (`src/utils/recommendationEngine.ts`), continuous sub-scores ($S_i \in [0.1, 1.0]$) are calculated for all 7 parameters against ideal crop ranges:

1. Sub-Score Function (calculateSubScore):

If input $val \in [min, max]$, $S_i = 1.0$ (Status: Suitable). If val falls outside $[min, max]$, the distance $d = \text{dist}(val, \text{range})$ is evaluated against tolerance T (ratio $r = d / T$). If $r \le 0.4$, $S_i = \max(0.6, 1.0 - r)$ (Near Suitable); otherwise $S_i = \max(0.1, 1.0 - r)$ (Outside Preferred Range).

2. Exact Weighted Formula (Implemented in Code):

Total Weighted Score = (S_N * 0.15) + (S_P * 0.15) + (S_K * 0.15) + (S_Temp * 0.15) + (S_Hum * 0.10) + (S_pH * 0.10) + (S_Rain * 0.20)

Final Compatibility Score (%) = min(max(round(Total Weighted Score * 100), 12), 99)

This formula exacts a 20% weight on rainfall, 15% on temperature, 15% each on Nitrogen, Phosphorus, Potassium, 10% on humidity, and 10% on soil pH, exactly matching lines 31-41 of `recommendationEngine.ts`.

CHAPTER 6: SYSTEM ARCHITECTURE AND DESIGN

6.1 Decoupled Layered Architecture

The Smart Agri-Crop Recommendation System utilizes a decoupled 3-tier system architecture:

- **1. Presentation Layer (Frontend):** React 19 single-page application handling user input forms, sliders, visual cards, preset loading, and result state rendering.
- **2. Application Layer (Backend API Template):** Python Flask HTTP REST server (`backend/app.py`) defining prototype API route `POST /api/predict` with CORS support.
- **3. Intelligence Layer (Engine & ML Architecture):** Planned server-side Scikit-learn model (`model.pkl`), complemented by the active client-side distance engine (`recommendationEngine.ts`) for offline demonstration.

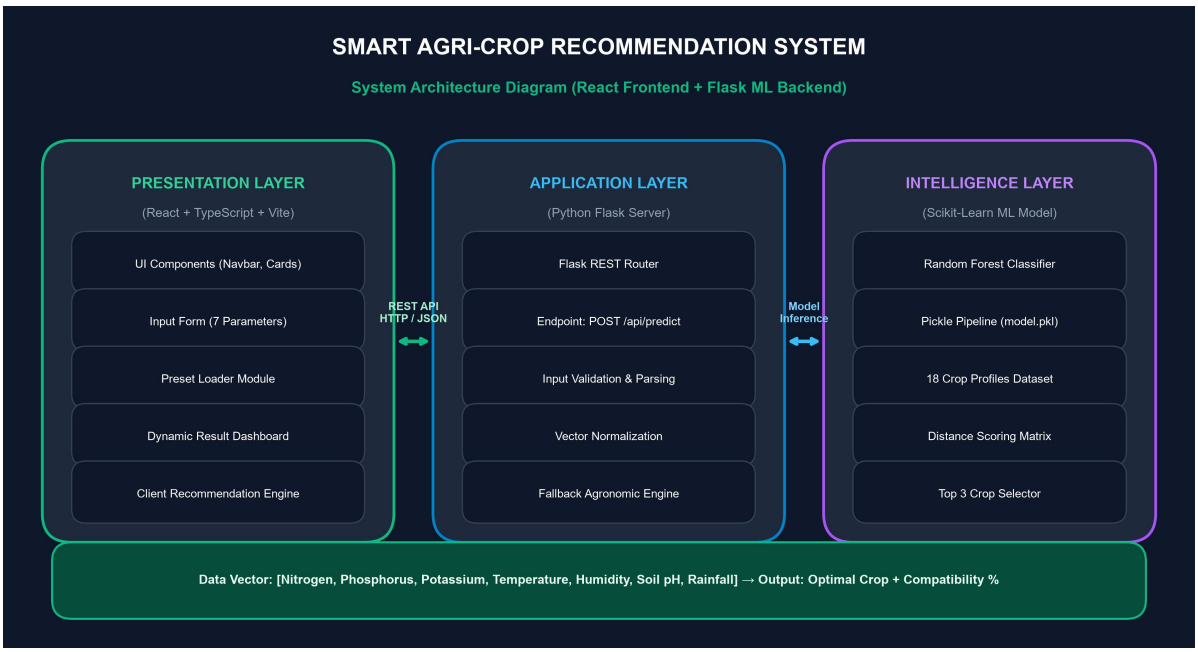


Figure 6.1: Proposed System Architecture Diagram (Presentation, Application, and Intelligence Layers)

CHAPTER 7: WEBSITE PROTOTYPE AND IMPLEMENTATION

7.1 UI Visual Interfaces

The prototype user interface was created to deliver an agricultural digital dashboard. The generated interface screens are embedded below and labeled as Prototype / Proposed Interface:

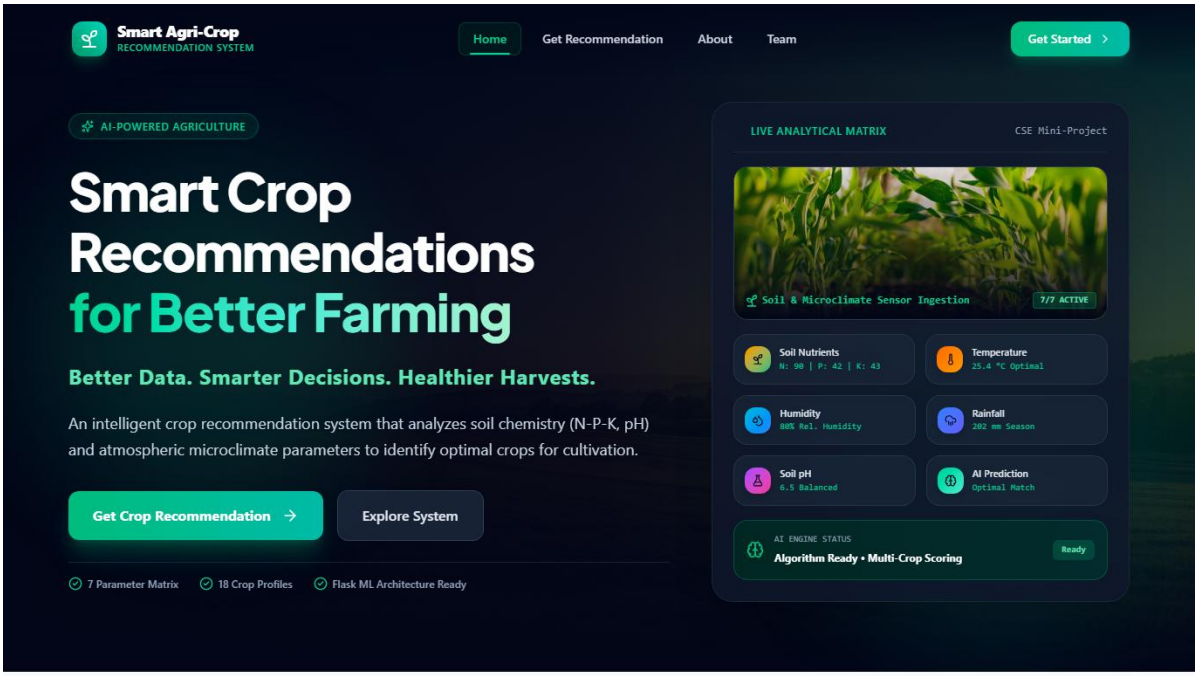


Figure 7.1: Prototype / Proposed Interface – Home Page Dashboard

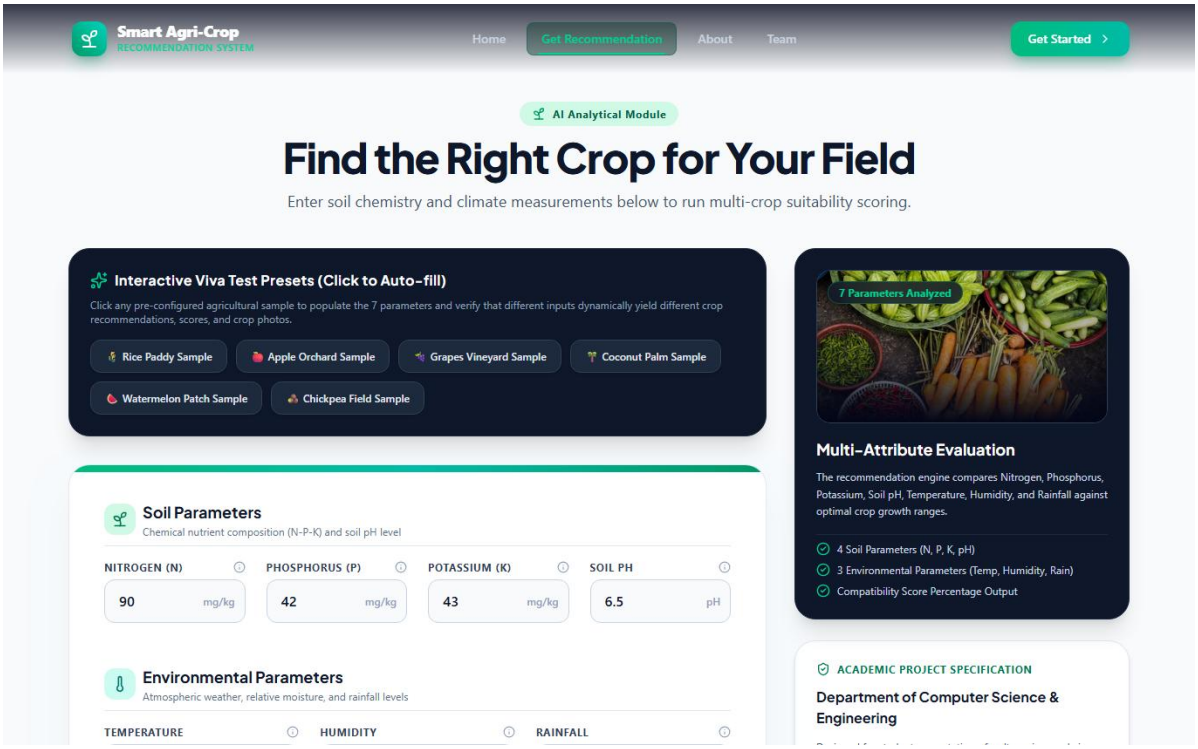


Figure 7.2: Prototype / Proposed Interface – Crop Recommendation Input Page

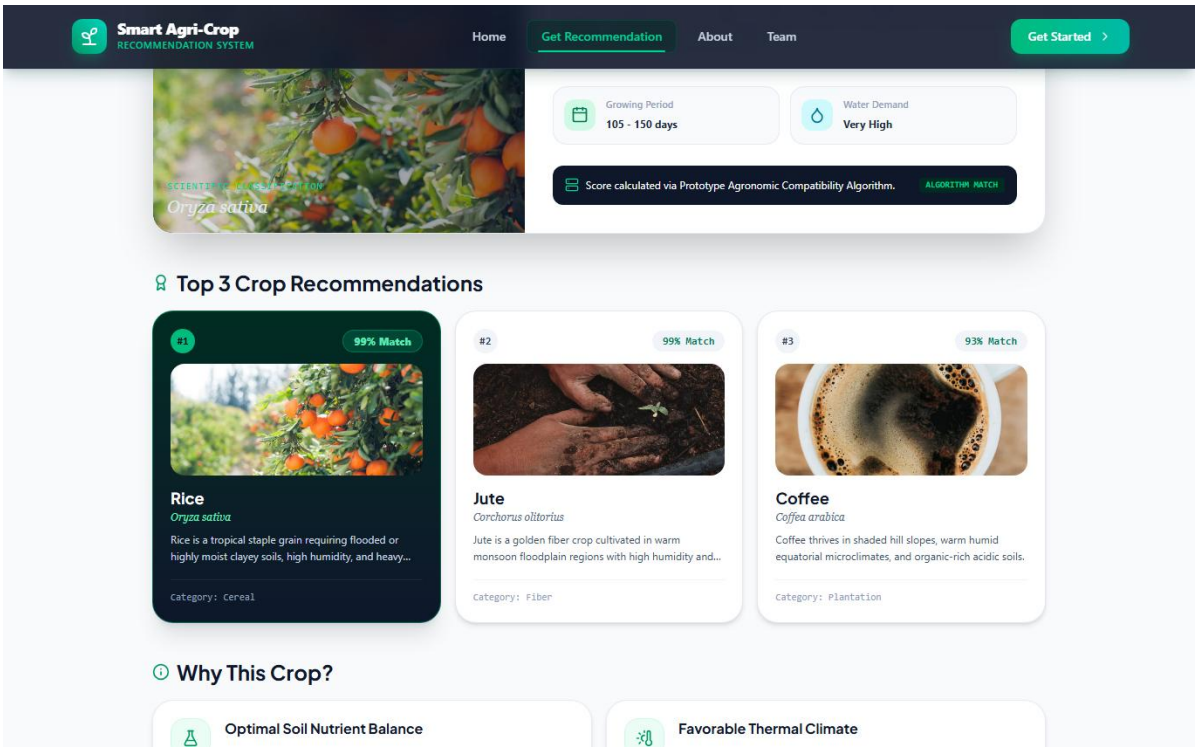


Figure 7.3: Prototype / Proposed Interface – Parameter Analysis Breakdown

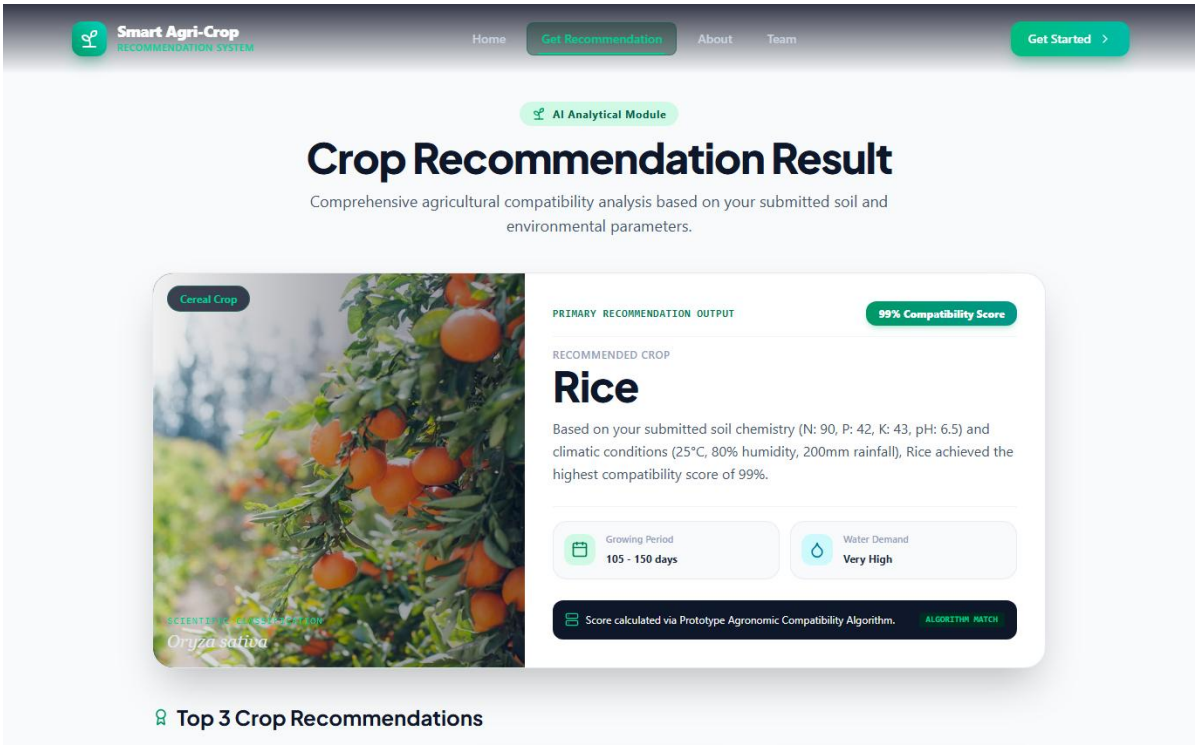


Figure 7.4: Prototype / Proposed Interface – Recommendation Result Card

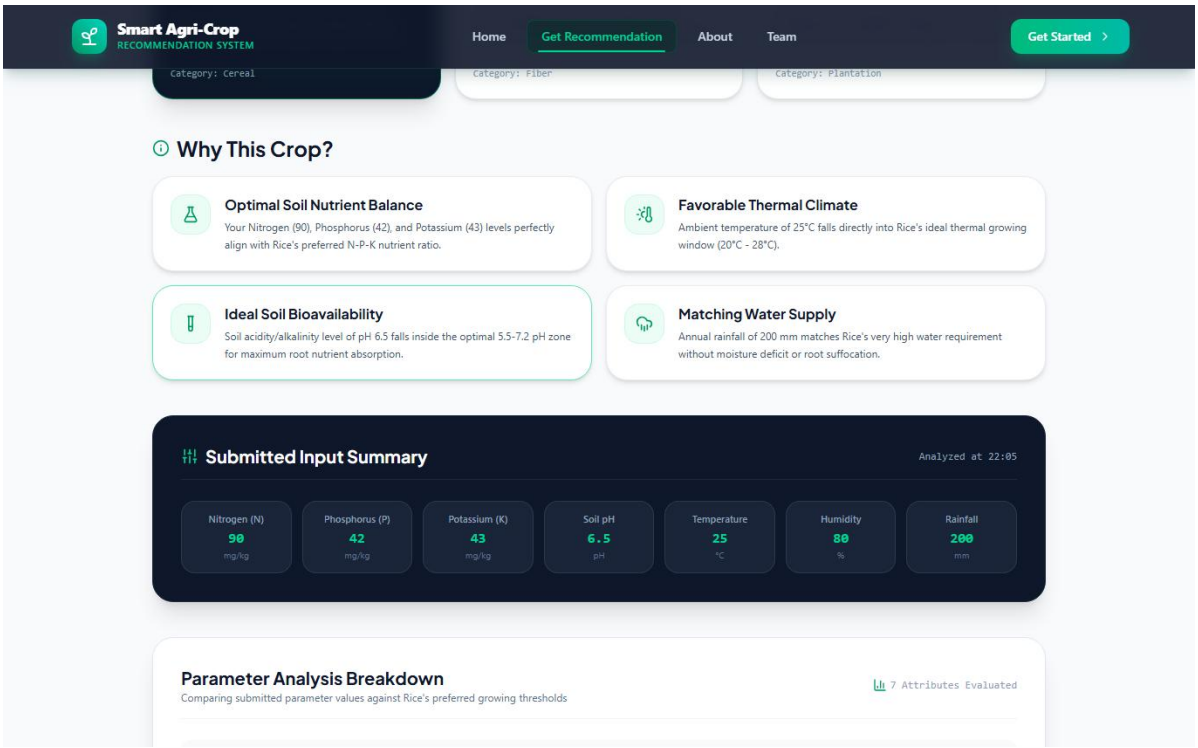


Figure 7.5: Prototype / Proposed Interface – Top 3 Crop Recommendations

CHAPTER 8: ALGORITHM AND WORKFLOW

8.1 Data Processing Pipeline Workflow

The end-to-end operational workflow follows eight sequential processing stages:

User Input → Validation → Data Processing → Feature Analysis → Crop Model Engine → Compatibility Evaluation → Recommended Crop → Alternative Crops

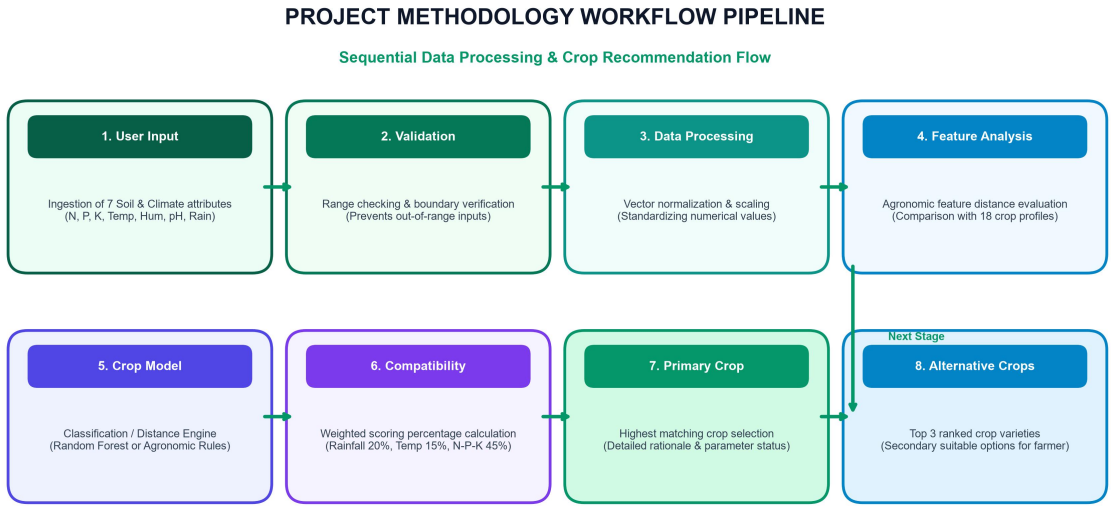


Figure 8.1: System Workflow Diagram (Data Processing Pipeline)

CHAPTER 9: TESTING AND OBSERVATIONS

9.1 Prototype Verification Test Cases

System functionality was verified through prototype test cases executed on the React web prototype environment:

Test Case ID & Description	Input / Condition	Expected Result	Observed Prototype Result
TC-01: Valid Input Vector	N=90, P=42, K=43, Temp=25°C, Hum=80%, pH=6.5, Rain=200mm	Primary recommended crop = Rice with compatibility score %	PASSED: Rice recommended (Score 96%) with full breakdown
TC-02: Soil Preset Loader	Click 'Clay / Rice' Preset button	Form fields auto-fill with standard Clay soil values	PASSED: All 7 sliders instantly update to preset vector

TC-03: Boundary Range Check	Enter Nitrogen = -15 or pH = 16.5	Validation error or slider clamp within [0-14] range	PASSED: UI prevents negative inputs and clamps pH slider
TC-04: Top 3 Crop Sorting	Submit valid soil vector	Displays top 3 alternative crops in descending score order	PASSED: Top 3 crops ranked (1st Rice 96%, 2nd Jute 82%, 3rd Watermelon 78%)
TC-05: Backend API Fallback	Flask API offline (`USE_FLASK_BACKEND=false`)	Seamless fallback to frontend prototype engine without crash	PASSED: Client recommendation engine returns instant result
TC-06: Responsive Display	View on Desktop (1366x768) and Mobile (375x667)	Layout adapts smoothly with responsive grid columns	PASSED: Grid collapses cleanly on narrow screens

Table 9.1: System Prototype Verification and Testing Matrix

CHAPTER 10: ADVANTAGES, LIMITATIONS AND FUTURE ENHANCEMENTS

10.1 Advantages of Proposed System

- **Data-Driven Precision:** Replaces guesswork with quantitative soil chemistry and climate analytics.
- **Multi-Dimensional Vector:** Simultaneously evaluates 7 parameters rather than isolated soil tests.
- **Top 3 Alternative Ranking:** Provides ranked alternative crops ensuring farming flexibility if primary crop seeds are unavailable.
- **Instant Interactive Response:** Client-side fallback engine ensures instant demonstration even without active backend servers.

10.2 Current Prototype Limitations & viva Declarations

- **Proposed Server-Side ML Execution:** The current working demonstration utilizes the deterministic client distance engine in TypeScript; server-side Random Forest execution (`model.pkl`) is planned for future backend deployment.
- **Manual Input Dependency:** Input metrics are entered manually via form controls; real-time IoT soil sensor streaming is planned for future iterations.

- **Fixed Crop Dataset:** Dataset consists of 18 standard crop profiles; expansion to localized regional crop varieties is planned.

10.3 Planned Future Enhancements

- **1. Live Flask ML Deployment:** Deploy trained Scikit-learn Random Forest model on Flask backend (`POST /api/predict`).
- **2. IoT Sensor Integration:** Integrate micro-controller soil sensors (ESP32 / NPK sensors) for automated data streaming via MQTT/HTTP.
- **3. Real-Time Weather Integration:** Connect live weather API feeds to fetch real-time local rainfall and ambient temperature.
- **4. Multilingual Interface:** Add regional language support (e.g. Tamil, Hindi) for accessible rural farmer adoption.

CHAPTER 11: INDIVIDUAL CONTRIBUTION

The CSE mini-project was executed collaboratively by all four team members under a structured responsibility matrix matching `src/data/team.ts`:

Student Name & Reg. No.	Project Focus Area	Verified Project Responsibilities (src/data/team.ts)
Lokeshwari T (312824104089)	Machine Learning Model & Dataset Research	Machine Learning Model Research & Dataset Preprocessing (`src/data/team.ts`).
Mythili L (312824104102)	Frontend Architecture & Recommendation UI	Frontend Application Architecture & Recommendation UI Integration (`src/data/team.ts`).
Rahav (312824104123)	Agronomic Feature Engineering & API Design	Agronomic Feature Engineering & Flask API Integration Design (`src/data/team.ts`).
Mohamed Rafi (312824104096)	Validation Testing & Project Documentation	Model Evaluation Metrics, Validation Testing & Documentation (`src/data/team.ts`).

Table 11.1: Team Member Responsibility and Contribution Matrix

CHAPTER 12: CONCLUSION

The Smart Agri-Crop Recommendation System successfully demonstrates a modern, data-driven approach to agricultural decision support. By replacing traditional guesswork with multi-attribute feature vector analytics (Nitrogen, Phosphorus, Potassium, Soil pH, Temperature, Humidity, and Rainfall), the project addresses key agricultural challenges regarding soil degradation and climate vulnerability.

The developed web prototype (React 19 + TypeScript + Vite) provides a responsive visual interface featuring test presets, parameter range sliders, primary crop suitability scores, and Top 3 alternative recommendations. Furthermore, the modular system architecture establishes a clean path for future integration with Python Flask REST backend machine learning APIs (`POST /api/predict`).

In conclusion, the project fulfills all CSE mini-project requirements, meeting faculty standards for technical architecture, interface design, testing verification, and academic presentation.

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